Projecting Wide Receiver Tiers With Machine Learning

By Duncan Park and Sameer Nathawat





Overview





01 **Background and Overview**

02 **Gaussian Mixture Model**

03 **Data collection & Preparation**

04 **Results & Conclusion**







Background

- Each year, undervalued WRs suddenly emerge as top performers
- What factors can be used to best predict wide receiver breakout seasons?



Collecting and Cleaning Data

- Weekly wide receiver data from 2012-2025 using the nflreadpy Python package
- Data included:
  - Yards, targets, TDs
  - Snaps played
  - Fantasy points

- Only wide receivers who played more than 20% of their teams offense snaps were included





Data Exploration



Feature Engineering

- Weekly wide receiver data was converted into → yearly wide receiver data
  - In order to not lose the complexity of the weekly data, yearly data was complied of statistics for early, mid, & late season

- Features such as yards/target, EPA/target, TD rate, etc. were created



Data Split





Gaussian Mixture Model



Results



Conclusion

